STEFANO BUDI

stefano.budi@binus.ac.id | +62 822 2980 9271 | stefano-budi.vercel.app | github.com/avalon-py | linkedin.com/in/stefano-budi

EDUCATION

BINUS University — *Jakarta, Indonesia*
B.S. in Computer Science, Data Science Specialization Track — *2024 – 2028 (Expected)*
GPA: 3.99 / 4.00
Independent Study: MIT 15.401, Harvard STAT110, Stanford CS229

PUBLICATIONS

"Dynamic Classifier Selection using Local Class Accuracy on an Overlap-Cleaned Heterogeneous Pool of Experts" — *Sept 2026*
2nd Author | Published

- Co-developed a dynamic classifier selection framework combining KNN-based overlap cleaning with local-class-accuracy-based expert selection across a heterogeneous pool of specialists (Naive Bayes, XGBoost, SVM), cutting log loss by 70.1% (0.898 → 0.269) versus a probability-harvesting baseline while achieving 91.62% accuracy and a macro F1-score of 0.8638 on the UCI Student Dropout dataset (n=4,424).

EXPERIENCE

Laboratory Assistant, BINUS University — *Feb 2025 – Feb 2026*

- Selected as 1 of 2 assistants from ~200 applicants (top <1%) through a 4-month, 3-stage process, including 1 month of full-time (7am–5pm) on-site bootcamp training, project creation, and presentation.
- Taught 5 classes weekly (~150 students) across Computer Vision and Algorithm & Programming (C), qualified in Deep Learning and Data Mining via BINUS's SLC qualifications; earned a 5.3/6 performance KPI.

PROJECTS

Speech2Market — End-to-End NLP-Based Market Impact Predictor for FED Speeches — *Aug 2026*
Dashboard | Live App | GitHub

- Engineered an NLP + time-series pipeline (classifier-less FinBERT embeddings, lagged feature engineering, macroeconomic indicators, HistGradientBoost) predicting SPX, Gold, VIX, and TNX returns at 3/7/30-day horizons following The FED's speeches.
- Deployed under a 1GB memory constraint via FP16 ONNX quantization of FinBERT, and built an automated GitHub Actions pipeline to work around FRED API/Yahoo Finance API blacklist, running entirely on free-tier infrastructure.

FindIT — Face Anti-Spoofing Detection — *Apr 2026*
Computer vision competition — camera-based identity fraud detection (DAC Find IT! 2026)

- Built a 5-fold DINOv2-based face anti-spoofing classifier across 6 classes using a two-phase fine-tuning strategy — frozen-backbone head training followed by full fine-tuning with differential learning rates — plus weighted sampling and selective MixUp to address class imbalance.
- Achieved 0.9742 macro F1, placing 11th of 340 teams (top 4%) within 0.005 of 1st place.

Laptop Price Predictor — End-to-End Ensemble ML Deployment — *May 2026*
Live App | GitHub

- Built a stacking ensemble (XGBoost, LightGBM, CatBoost → RidgeCV meta-learner) tuned via Optuna Bayesian optimization, achieving 6.67% MAPE and outperforming all base models in accuracy and cross-validation stability.
- Shipped a full pipeline from training with asynchronous web-scraped dataset (BeautifulSoup + Playwright, 583 listings) to a deployed Streamlit app (P90 latency ~65–75ms), validated via a 5-user study with 100% "useful" rating.

MBG Investment Priority Mapping — Graph-Based Clustering Research Project — *Jul 2025*
Dashboard | GitHub

- Applied spectral (graph-based) clustering with a Laplacian structure to map investment priority across 38 Indonesian provinces, engineering 15+ derived features from raw government data (BPS, Badan Pangan Nasional), filtered via RFECV, validated via the Calinski-Harabasz index (3.94 vs. a 1.01 random-baseline benchmark); selected by BINUS's Computer Science faculty as product presentation material for a public open event.

TECHNICAL SKILLS

Languages: Python, SQL (fluent); C (proficient); Java (basic); RapidMiner, R (familiar)
Machine Learning: Gradient Boosting, Spectral Clustering, Temporal Fusion Transformers, Naïve Bayes, Decision Trees, SVM, PCA/LDA, Optuna
Transfer Learning: DINOv2, ConvNeXt, FinBERT, ONNX Runtime, LLM Application Integration
Data Engineering & Deployment: Pandas, BeautifulSoup, Playwright, GitHub Actions, Streamlit, Git/GitHub